```
222     def _init_db(self):
223         cur = self._conn.cursor()
224         cur.execute("PRAGMA journal_mode=WAL")
225         cur.execute("PRAGMA synchronous=NORMAL")
226         cur.execute("PRAGMA temp_store=MEMORY")
227         cur.execute(
228             """
229             CREATE TABLE IF NOT EXISTS flight_latest (
230                 flight_key TEXT PRIMARY KEY,
231                 icao_address TEXT NOT NULL,
232                 callsign TEXT NOT NULL,
233                 timestamp TEXT NOT NULL,
234                 lat_e6 INTEGER NOT NULL,
235                 lon_e6 INTEGER NOT NULL,
236                 altitude_m INTEGER NOT NULL,
237                 speed_cms INTEGER NOT NULL,
238                 heading_de2 INTEGER NOT NULL,
239                 vs_cms INTEGER NOT NULL,
240                 squawk INTEGER,
241                 source_ip TEXT,
242                 sample_count INTEGER NOT NULL DEFAULT 0
243             )
244             """
245         )
246         cur.execute(
247             """
248             CREATE TABLE IF NOT EXISTS trajectory_samples (
249                 id INTEGER PRIMARY KEY AUTOINCREMENT,
250                 flight_key TEXT NOT NULL,
251                 timestamp TEXT NOT NULL,
252                 lat_e6 INTEGER NOT NULL,
253                 lon_e6 INTEGER NOT NULL,
254                 altitude_m INTEGER NOT NULL,
255                 speed_cms INTEGER NOT NULL,
256                 heading_de2 INTEGER NOT NULL,
257                 vs_cms INTEGER NOT NULL,
258                 squawk INTEGER,
259                 source_ip TEXT,
260                 UNIQUE(flight_key, timestamp)
261             )
262             """
263         )
264         cur.execute(
265             "CREATE INDEX IF NOT EXISTS idx_traj_flight_time ON trajectory_samples(flight_key, timestamp)"
266         )
267         self._migrate_lat_precision_schema(cur)
268         self._refresh_schema_flags(cur)
269         self._conn.commit()
270 
271     def reset_for_new_session(self):
272         """
273         Wipe trajectory/prediction tables and in-memory caches.
274         Called at startup in simulation mode so stale flight paths from a
275         previous run can never be mixed with the current run's timestamps
276         (which would make the predictor extrapolate from the wrong segment
277         of a route and appear to "trace back").
278         """
279         if not self.enabled:
280             return
281         with self._lock:
282             cur = self._conn.cursor()
283             cur.execute("DELETE FROM trajectory_samples")
284             cur.execute("DELETE FROM flight_latest")
285             self._conn.commit()
286             self._last_saved.clear()
287             self._last_ingest_wallclock.clear()
288             self._sample_counter.clear()
289 
290     def _table_columns(self, cur, table_name):
291         cur.execute(f"PRAGMA table_info({table_name})")
292         return {row[1] for row in cur.fetchall()}
293 
294     def _migrate_lat_precision_schema(self, cur):
```